

Literature review UUV

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1 Introduction

Unmanned Underwater Vehicles (UUVs) serve as critical equipment capable of executing prolonged, long-range autonomous operations within complex marine environments. They hold extensive application prospects in fields such as deep-sea resource exploration, seabed topography mapping, marine environmental monitoring, maritime security assurance, and military reconnaissance [1, 2]. In recent years, with the continuous advancement of marine development strategies, the importance and demand for UUVs have exhibited a marked upward trend. In practical applications, autonomous navigation and obstacle avoidance capabilities constitute one of the core technologies for UUVs to successfully complete missions [3]. Ideally, a UUV should be capable of autonomously planning safe, feasible paths through unknown or partially known underwater environments, travelling from a starting point to a target location while effectively avoiding both static obstacles (such as seabed topography and shipwreck debris) and dynamic obstacles (such as other vessels, marine life, and floating objects) [4]. Furthermore, the underwater environment presents significant internal and external disturbances, including turbulence caused by currents and waves, acoustic sensor noise, and propeller thrust fluctuations. These stochastic factors impose significant challenges on the stability and precision of trajectory execution in the vehicle, further complicating path planning and trajectory tracking [5]. However, despite extensive research in the field of path planning and control algorithms for unmanned underwater vehicles, Existing literature frequently decouples high-level path planning from low-level motion control—such as high-level planning or low-level control—without adequately considering their interactions within a unified system framework. Particularly under real-world dynamic disturbance conditions, the gap between planned paths and executable trajectories remains a critical challenge. Furthermore, the absence of flexible, reproducible simulation platforms capable of synergistically modelling vehicle dynamics, environmental disturbances, and navigation algorithms constrains the effective validation and comparison of diverse methodologies. Therefore, there is an urgent need to construct a realistic simulation environment integrating the motion dynamics of underwater unmanned vehicles, dense obstacle environments, and closed-loop path planning and trajectory tracking strategies. Such simulators can serve as vital tools for evaluating the feasibility, robustness, and safety of navigation algorithms under controlled and representative underwater conditions, thereby supporting both academic research and engineering practice applications.

2 UUV Kinematics and Dynamics Modeling

2.1 The Evolution from 3-DOF to 6-DOF

Unmanned Underwater Vehicles (UUVs) are typically described using 3-DoF or 6-DoF nonlinear dynamic models. The 3-DoF model is commonly employed for planar (2D) motion analysis, retaining only surge (u), sway (v), and yaw (r). It features lower mathematical complexity with minimal nonlinear coupling terms, and the inertia matrix is 3×3 . The pose vector is defined as:

$$\boldsymbol{\eta} = [x \quad y \quad \psi]^T \dots (1)$$

This model is suitable for constant-depth cruising scenarios where obstacle avoidance is limited to the horizontal plane, and may suffice for pure path planning simulations. However, it lacks capability in vertical avoidance and coupled attitude motions, making it insufficient for integrated testing of both path planning

and obstacle avoidance control algorithms. Therefore, this project adopts the 6-DoF full-space model. The 6-DoF model captures the full coupling effects of translational and rotational motions in three-dimensional space, typically incorporating rigid-body dynamics, hydrodynamic effects, and external forces/moments acting on the vehicle. It enables complex maneuvers such as vertical obstacle avoidance, spiraling ascents, and significant attitude changes. However, it involves substantially higher mathematical complexity, with a 6×6 inertia matrix and intricate trigonometric coupling terms. The pose vector is defined as:

$$\boldsymbol{\eta} = [x \quad y \quad z \quad \phi \quad \theta \quad \psi]^T \dots (2)$$

2.2 Under-actuated and Constrained Systems

Building upon the foundation of 6-DOF models, researchers often simplify and decouple these models according to specific task requirements. In deep-sea long-range cruising or specific planar obstacle avoidance tasks, UUVs are typically treated as systems moving within a defined plane. For instance, to address depth transition and pitch control, researchers frequently extract a dive plane model, simplifying the 6-DOF system into a 3-DOF subsystem comprising forward velocity (u), vertical velocity (w), and pitch angular velocity (q).

However, this simplification introduces significant underactuation, where the system's control input dimensions are fewer than its degrees of freedom. In typical torpedo-shaped UUVs, actuators are typically limited to a tail thruster and rudder surfaces, whose dynamic constraints can be expressed as:

$$\begin{cases} m_{11}\dot{u} = F_u(u, w, q) + \tau_u \dots (3a) \\ m_{33}\dot{q} = F_q(u, w, q) + \tau_q \dots (3b) \\ m_{22}\dot{w} = F_w(u, w, q) \dots (3c) \end{cases}$$

where τ_u represents forward thrust, τ_q denotes pitch moment, and the vertical degree of freedom w lacks direct control input (i.e., $\tau_w = 0$). This implies that depth changes must be achieved indirectly by adjusting the pitch angle via τ_q , leveraging hydrodynamic coupling effects. This physical constraint necessitates control algorithms capable of handling non-holonomic constraints. Additionally, motion constraints manifest in the physical saturation of actuators, such as thrust upper bounds $\tau \leq \tau_{max}$ and pitch angle range constraints $\theta \in [-\pi/2, \pi/2]$ to avoid mathematical singularities. In complex marine environments, ensuring the stability and reachability of control systems under these constraints represents a core challenge in current UUV simulation and control research.

2.3 Kinematic and Dynamic Models

Similar to other robotic systems, UUV navigation control strategies can be classified into kinematics-based and dynamics-based approaches. Kinematics-based methods primarily focus on the geometric relationship between position and velocity, often assuming idealized actuation and neglecting the vehicle's dynamic characteristics. For example, typical kinematic equations include [6]:

$$\begin{aligned} \dot{x} &= u \cos \theta + w \sin \theta, \dots (4a) \\ \dot{z} &= -u \sin \theta + w \cos \theta, \dots (4b) \\ \dot{\theta} &= q, \dots (4c) \end{aligned} \tag{3}$$

More generally, the kinematic behavior of UUVs is described by a nonlinear transformation equation that relates body-frame velocities to inertial-frame position and orientation:

$$\dot{\boldsymbol{\eta}} = \mathbf{J}(\boldsymbol{\eta})\boldsymbol{\nu} \dots (5) \tag{4}$$

where $\boldsymbol{\nu}$ denotes the body-fixed velocity vector and $\mathbf{J}(\boldsymbol{\eta})$ is the nonlinear coordinate transformation matrix. Such formulations accurately capture the nonholonomic characteristics of underwater vehicles and are widely used in trajectory generation and guidance design. In contrast, dynamics-based control methods explicitly account for physical factors such as mass distribution, hydrodynamic damping, added mass, and external

forces, making them more suitable for real-world underwater operations. The dynamic behavior of UUVs is typically modeled using the following compact nonlinear equation:

$$\mathbf{M}(\boldsymbol{\nu})\dot{\boldsymbol{\nu}} + \mathbf{C}(\boldsymbol{\nu})\boldsymbol{\nu} + \mathbf{D}(\boldsymbol{\nu})\boldsymbol{\nu} + \mathbf{g}(\boldsymbol{\eta}) = \boldsymbol{\tau} \dots (6)$$

where $\mathbf{M}(\boldsymbol{\nu})$ is the inertia matrix (including rigid-body and added mass), $\mathbf{C}(\boldsymbol{\nu})$ is the Coriolis and centripetal matrix, $\mathbf{D}(\boldsymbol{\nu})$ is the hydrodynamic damping matrix, $\mathbf{g}(\boldsymbol{\eta})$ is the vector of restoring forces and moments due to gravity and buoyancy, and $\boldsymbol{\tau}$ represents the control input (generalized forces/moments).

2.4 Modeling Complex Oceanic Disturbance Environments

Disturbances in underwater environments (such as waves and currents) are commonly modeled using stochastic processes (e.g., Gaussian noise) or periodic functions to represent wave-induced motions. Although some studies attempt precise estimation and modeling of these disturbances, recent trends favor control strategies with inherent robustness—treating disturbances as external uncertainties to be compensated via feedback and online optimization, rather than relying on highly accurate a priori modeling [5].

3 Review of UUV Path Planning Algorithms

3.1 Problem Formulation and Planning Paradigms

Path planning for UUVs is broadly classified into two categories: point-to-point path planning and complete coverage path planning. The former seeks a path from a start point A to a goal point B, while the latter requires traversing all reachable positions in a designated area (e.g., search missions). Point-to-point planning is further divided into global offline planning, which computes the entire route using known map information before departure, offering optimality but inability to handle sudden changes; and local dynamic planning, which adjusts the path in real-time based on surrounding environment, suitable for unknown or dynamic settings but prone to local optima and lacking global perspective [7].

3.2 Global Path Planning Methods for UUV Navigation

Among global planning algorithms, the heuristic search method A* is widely used. It evaluates the eight neighboring grids of the current position and selects the next step with the lowest cost via a heuristic function—the smaller the heuristic value, the higher the likelihood of being on the optimal path. However, traditional A* has notable limitations in UUV navigation: it cannot guarantee global optimality in the presence of multiple local minima; moreover, its fixed turning angles (typically restricted to eight directions, $\geq 45^\circ$) result in excessive waypoints and turns, adversely affecting UUV maneuverability and energy consumption [1]. Ant Colony Optimization (ACO) is another prevalent global planning approach. It mimics ant foraging behavior by randomly deploying artificial “ants” and selecting the next node based on a transition probability, which primarily depends on pheromone concentration $\tau_{rs}(t)$ and heuristic information, such as $\eta_{rs} = \frac{1}{d_{rs}}$. Ant Colony Optimization (ACO) exhibits strong adaptability and high parallelism, making it suitable for various combinatorial optimization problems [7]. Zhi-Wen et al. proposed an enhanced approach by integrating ACO with a biological-inspired neural network for global planning. This method incorporates pheromone concentration, neural activity value ($f_{rs}(t)$, with higher attraction to unvisited grids), distance heuristic, and turning angle factor ($A_{rs}(t)$, to minimize turns). The transition probability is given by:

$$P_{rs}^k(t) = \frac{\tau_{rs}(t) f_{rs}(t) \eta_{rs} A_{rs}}{\sum_x \tau_{rx}(t) f_{rx}(t) \eta_{rx} A_{rx}} \dots (7)$$

By multiplying these four factors to compute the integrated probability, the method selects the statistically best path on average, effectively reducing the number of turns and energy consumption for UUVs [7]. In local planning, the Artificial Potential Field (APF) method is popular due to its low computational demand and strong real-time performance. It models the environment as an artificial potential field where the goal exerts an attractive force on the UUV, while obstacles exert repulsive forces, enabling the vehicle to move under the resultant force for safe path generation. Nevertheless, APF suffers from inherent issues: when

attractive and repulsive forces are equal in magnitude but opposite in direction, the resultant force becomes zero, causing stagnation or oscillation; additionally, excessive repulsion near obstacles close to the goal may prevent reaching the target [1].

3.3 Enhanced and Hybrid Global Planning Strategies

To address the limitations of individual algorithms, many studies adopt hybrid architectures combining global and local planning. Zhi-Wen et al. proposed a representative strategy: first generate a coarse global path using the improved ACO, then insert guide points (Guide Point Strategy) along the path to decompose the problem into multiple smaller local sub-problems. Within each sub-problem, a rolling window mechanism is employed, performing real-time planning only within the local sonar-detected region. The method also prioritizes avoidance behaviors: heading change is preferred (maintaining constant speed, only adjusting direction) due to its flexibility and energy efficiency; speed reduction or stopping is used only when necessary [7].

3.4 Local Path Planning and Real-Time Obstacle Avoidance

In contrast, Wu et al. employed a different hybrid approach: using APF as the primary local planner throughout, while monitoring real-time displacement to detect trapping in local minima. Upon detection of a local minimum, an escape mechanism is triggered—temporarily switching to A* to find a path out of the “dead zone,” then resuming APF tracking. This strategy effectively mitigates APF’s local minimum issue while preserving its efficient real-time obstacle avoidance capability. Although A* or APF+A* algorithms can effectively generate globally collision-free paths, their outputs typically consist of discrete geometric waypoints that lack consideration for UUV dynamic constraints and current disturbances. Therefore, this paper utilizes these waypoints as reference inputs within a Model Predictive Control (MPC) framework to perform local trajectory optimization and smoothing.

4 Trajectory Tracking Control Strategy

4.1 Challenges in UUV Trajectory Tracking

After completing path planning, the core challenge for navigation systems lies in precisely driving the UUV to follow a predetermined trajectory within complex, disturbance-ridden underwater environments. Traditional PID control methods often struggle to maintain system robustness during high-speed navigation or under severe environmental disturbances due to the nonlinear and non-regular behavior of UUVs. [5]

4.2 Robust Control via Sliding Mode Techniques

Addressing these challenges, Sliding Mode Control (SMC) has gained significant academic attention due to its exceptional robustness against model parameter uncertainties and external disturbances. Researchers often construct nonlinear sliding surfaces to force the system state to evolve along the intended trajectory. For instance, in trajectory tracking along the dive plane for underactuated UUVs, specific switching functions can effectively mitigate the impact of modeling errors [6]. However, the “chattering” phenomenon inherent in traditional SMC may cause physical wear on thrusters and struggles to directly handle the physical limit constraints of actuators.

4.3 Constraint-Aware Trajectory Tracking Using MPC

In recent years, Model Predictive Control (MPC) has emerged as a mainstream solution in trajectory tracking due to its unique predictive mechanism and constraint handling capabilities. Unlike traditional feedback control, MPC leverages the UUV’s 6-DOF kinematic and dynamic models to perform rolling optimization calculations within a finite time domain [2]. The core advantage of this mechanism lies in its explicit handling of complex motion constraints. As highlighted in relevant studies, directly incorporating thrust saturation and attitude singularity constraints (e.g., $|\theta| < \pi/2$) into the Quadratic Program (QP) optimization problem

yields control commands that are both physically feasible and energy-efficient [5]. To further enhance the system’s adaptive capabilities in dynamic environments, some researchers have explored integrating intelligent optimization algorithms with traditional controllers. For instance, fuzzy logic is employed to adjust control gains in real time, or reinforcement learning is utilized to directly learn optimal action strategies within continuous control spaces, thereby improving the UUV’s resistance to sonar noise and unknown currents [7, 8].

5 Summary and Research Motivation

In summary, current research on unmanned underwater vehicle navigation has achieved significant progress in three key areas: high-fidelity kinematic and dynamic modeling, efficient global and local path planning algorithms, and robust trajectory tracking control strategies. Six-degree-of-freedom dynamic models provide a solid theoretical foundation for representing real-world motion, while hybrid path planning frameworks effectively integrate global optimality with local adaptability. Concurrently, advanced control techniques such as sliding mode control and model predictive control demonstrate robust disturbance rejection against nonlinearities, disturbances, and actuator constraints.

Nevertheless, several critical limitations persist: most studies treat path planning and trajectory tracking as decoupled modules, resulting in planned paths that are difficult to execute fully under dynamic constraints and environmental disturbances. Furthermore, some studies generate purely geometric waypoints in planning algorithms without explicitly accounting for underactuation, thrust saturation, and ocean current effects.

These gaps highlight the necessity of constructing an integrated simulation framework that tightly couples global-local path planning with constraint-aware trajectory optimization and tracking control. This enables fair comparison and systematic validation of navigation strategies. To address this need, this study focuses on developing a realistic simulation environment for unmanned underwater vehicles (UUVs). This environment incorporates six-degree-of-freedom dynamics, random ocean disturbances, hybrid path planning algorithms, and model predictive control-based trajectory tracking technology. It provides a comprehensive testing platform for evaluating navigation performance in complex underwater scenarios.

References

- [1] X. Wu, X. Long, S. Yuan, Q. Hu, and P. Xie, “Multi-uuv coordinated path planning with collision avoidance based on combination of improved apf and a*,” in *2022 8th International Conference on Control, Automation and Robotics (ICCAR)*, Xiamen, China, 2022, pp. 218–223.
- [2] S. Huh, “Unmanned underwater vehicle (uuv) systems for submarine detection: A technology primer,” 2019. [Online]. Available: <https://ontheradar.csis.org/issue-briefs/unmanned-underwater-vehicle-uuv-systems-for-submarine-detection-a-technology-primer/>
- [3] B. D. Richard, “The development of autonomous underwater vehicles (auv): A brief summary,” in *Proceedings of the IEEE International Conference on Robotics and Automation*, Seoul, Korea, 2001.
- [4] L. YongZhou, L. GuangYu, and G. Xuan, “Multi-uuv path planning study with improved ant colony algorithm and ddqn algorithm,” in *2021 IEEE 7th International Conference on Control Science and Systems Engineering (ICCSSE)*, Qingdao, China, 2021, pp. 143–148.
- [5] Y. Zhang, X. Liu, M. Luo, and C. Yang, “Mpc-based 3-d trajectory tracking for an autonomous underwater vehicle with constraints in complex ocean environments,” *Ocean Engineering*, vol. 189, p. 106309, 2019.
- [6] Z. Yan, H. Yu, B. Li, and J. Zhou, “Sliding mode trajectory tracking of underactuated uuv on dive plane,” in *Proceedings of the 33rd Chinese Control Conference*, Nanjing, China, 2014, pp. 7909–7914.
- [7] Z.-W. Wang, K. Li, and L.-J. Wang, “Path planning for uuv in dynamic environment,” in *2016 9th International Symposium on Computational Intelligence and Design (ISCID)*, Hangzhou, China, 2016, pp. 211–215.

- [8] F. Wang, K. Qian, R. Zhang, and L. Jiang, “Application of qpso algorithm in aeroengine performance optimization,” in *2021 IEEE 2nd International Conference on Information Technology, Big Data and Artificial Intelligence (ICIBA)*, Chongqing, China, 2021, pp. 1–4.